Institut Polytechnique
Des Sciences Avancées

81 Av. de Grande Bretagne
31300 Toulouse

Laboratoire de Mécanique
des Fluides de Lille

8 Bd Louis XIV
59000 Lille

École Nationale Supérieure
des Arts et Métiers

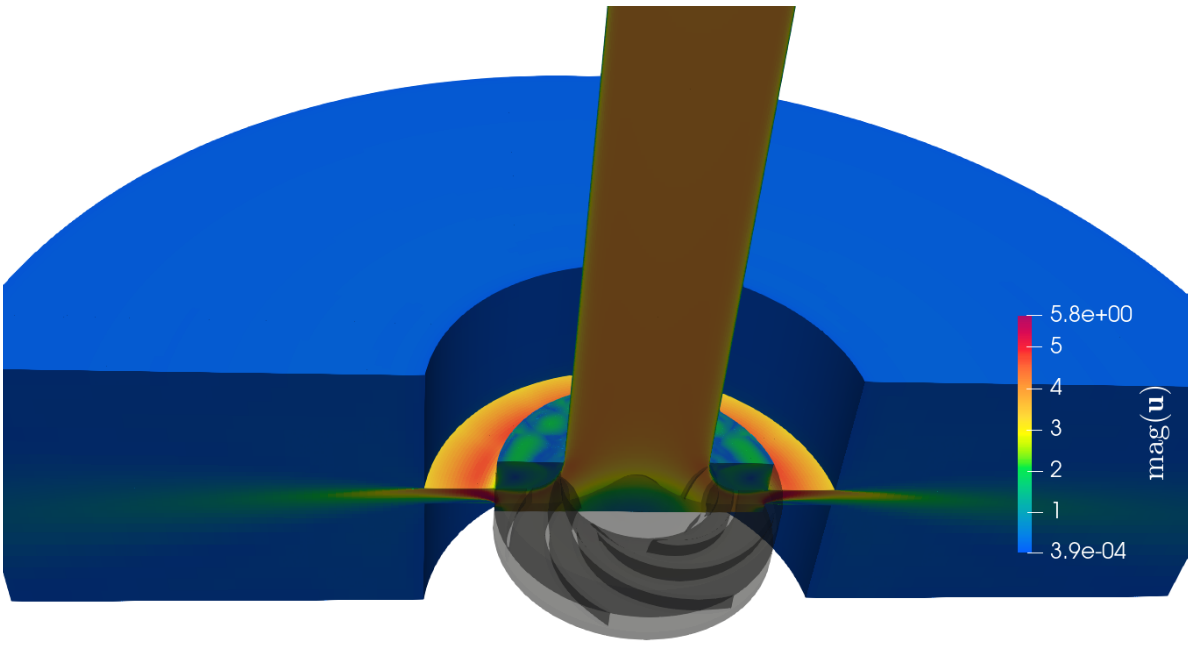
8 Bd Louis XIV
59000 Lille

## End-of-study thesis for engineering degree

---

# Optimization of the performance of a centrifugal pump using Data Assimilation

## Intern in Rotating Machines




**Cross-section of the pump model**

Educational tutor: Samy LABSIR
Laboratory tutors: Marcello MELDI, Antoine DAZIN, Miguel MARTINEZ VALERO

Clément THIBAULT - Aéro 5

From February $26^{th}$, 2025 to September $2^{nd}$, 2025

Version of July 17, 2025

# Acknoledgments

## Acknowledgements

First and foremost, I would like to sincerely thank my three internship supervisors. I am deeply grateful to Miguel Valero Martinez for his continuous support, patience, and guidance throughout this internship. His expertise and encouragement have been truly invaluable and will continue to inspire me throughout my thesis and beyond. Many thanks as well to Marcello Meldi for welcoming me into the LMFL and for his insightful advice at every step. I would also like to warmly thank Antoine Dazin, whose clear physical explanations helped me grasp the key concepts behind the experimental pump.

I am especially grateful to Tom, my office mate, for his generosity and clarity in explaining complex topics. His help greatly deepened my understanding on many subjects. Thanks to Paolo for his valuable support and for introducing me to CONES. To Sarp, thank you for your energy and positivity, which made the lab a very enjoyable place to work.

I'd also like to thank my LMFL colleagues Bo, Eddy, Francis, Vincent, Jeanne, Thibaut, and Ernesto for their kindness and camaraderie throughout this experience. Many thanks as well to my friends Dimitri, Benjamin, and Joan for their help with physics, mathematics, and numerical methods.

I am also grateful to the administration teams at ENSAM and LMFL for their support, and to the TGCC for providing access to HPC resources. My thanks to the ANR-JCJC (Agence Nationale de la Recherche — Jeunes Chercheuses et Jeunes Chercheurs), specifically ANR-JCJC-2021-IWP-IBM-DA, which funded this internship.

I warmly thank my family and friends for their constant encouragement throughout my studies and this internship. Their belief in me has been a constant source of motivation.

Last but not least, thank you to Samy LABSIR for agreeing to be my IPSA tutor for this internship and for always being available when needed.

# Contents

## Summary sheet

The aim of this technical assessment is to summarize the gaps and their causes between the course's missions and what has been achieved. It also enables us to suggest what could be done by the next person working on the same subject.

| **Summary sheet** | Clément THIBAULT - Aéro 5 TIE |
|---|---|
| **Internship topic** | **Objectives** |
| Optimization of the performance of a centrifugal pump using Data Assimilation | Optimize the forcing parameters of an IBM[1] in a centrifugal pump using DA with CONES, to find the correct performance of the pump. |
| **Main customer** | **Used tools** |
| - LMFL<br>- ENSAM | OpenFoam 9, CONES, VSCode, LaTeX, Git, Python,C++, MPI, Bash, TGCC, Paraview |
| **Conducted studies** | |
| - Coding the DA test case in OpenFoam and CONES<br>- Convergence of the CFD solution (physical and numerical stability/precision)<br>- Convergence of the EnFK, adding tools as inflation | |
| **Results** | **Explanation of possible differences** |
| - Drop up to 50% of the mathematical NRMSE with DA (almost 100% expected)<br>- Implementation of inflation, in CONES<br>- Implementation of radial expansion, "azimutal interpolation" in OpenFoam 9<br>- Discovering of a recirculation bubble in the diffuser | - After the diffuser, the impeller's wake direction is reversed because of the MRF method<br>- The rotating impeller shape combined with the IBM allows non-physical flow that induce a recirculation bubble.<br>- Results of the EnKF are not as good as expected because of the IBM method combined to MRF method, which is not adapted to the centrifugal pump geometry. |
| **Encountered difficulties** | **Work to be continued** |
| - Understanding the tools OpenFoam and CONES<br>- Code everything directly on the TGCC terminal<br>- Initialize physical field of complex instationary simulation | - Use the same type of DA for the analysis of axial compressor internal flows<br>- Synchronize PIV and simulation in the DA<br>- Implement interpolation IBM method |

**Note:** Acronymes are define later.

[1] All acronyms are defined from the next page or available in the glossary.

# General Introduction

Since I was young, I've always been fascinated by fluid mechanics. I remember watching planes flying and being amazed by the way they could fly. I was also interested in the way water flows in rivers and sailing boats. In high school, I began working on plane-related projects and explored them more theoretically through Computation Fluid Dynamics (CFD).

This fascination led me to study physics and mathematics at university, where I discovered applied mathematics including CFD, Artificial Intelligence (AI) and Kalman Filter. During my studies, I've become passionated about applied mathematics and fluid mechanics. That's why I did my last internship at the Centre Européen de Recherche et de Formation Avancée en Calcul Scientifique (CERFACS). It convinced me to do a thesis in CFD after Institut Polytechnique des Sciences Avancées (IPSA), my current engeneering school.

The subject proposed by the Laboratoire de Mécanique des Fluides de Lille (LMFL) really suited me: CFD using an open-source library, coupled with AI and experimental data. So it was with enthusiasm that I started this internship on February 26, 2025, in this laboratory for a duration of six months.

Briefly, the LMFL is a public laboratory, consisting of 38 researchers and around 25 PhD students. The laboratory is present on three sites located in Lille (École Nationale Nupérieure d'Arts et Métiers (ENSAM) and Office National d'Études et de Recherche Aérospatiales (ONERA)) and on the Villeneuve d'Ascq campus (Centrale Lille, Centre National de la Recherche Scientifique (CNRS), University of Lille).

My main tasks during this internship were:

- Ensuring the convergence of the CFD solution;
- Implementing the Data Assimilation (DA) problem in Open source Field Operation And Manipulation (OpenFOAM) and Coupling OpenFoam with Numerical EnvironmentS (CONES);
- Optimizing the forcing parameters of the Immersed Boundary Methods (IBM) using DA with CONES to correct a non-physical behavior.

To present the details of my internship work, I will first introduce the LMFL research laboratory. In the second part, I will describe the work I carried out, structured into the following five sections:

1. Introduction to DA problems;
2. Numerical ingredients of the CFD and DA problems;
3. Preliminary results using IBM;
4. Application of DA to the IBM;
5. Results and perspectives.

# Chapter 1

# The LMFL laboratory and Arts et Métiers Lille campus

## 1.1 Scientific environment and host institution

The LMFL is a public research laboratory jointly supervised by Centrale Lille, Université de Lille, Arts et Métiers ParisTech, ONERA, and the CNRS.

Figure 1.1: LMFL facilities; Top:Villeneuve d'Ascq campus, bringing together Centrale Lille, the CNRS, and Université de Lille. Bottom-left: ONERA. Bottom-right: Arts et Métiers campus (ENSAM) (source : lmfl.cnrs.fr)

Over the past six months, I have been hosted by ENSAM, specifically at the Lille campus. Arts et Métiers is a prestigious engineering school focused on mechanical, industrial, and energy engineering. The Lille campus to students a dynamic and innovative environment in a region with a strong industrial heritage.

The LMFL was created on January 1, 2018, from the merger of two research units: the "Écoulements Tournants et Turbulents" team from the Lille Mechanics Laboratory, and the "Expérimentation et Limite de Vol" unit. This diversity of expertise enables the research teams to engage in numerous collaborative projects with regional and national partners.

Research conducted within the LMFL is organized into three main topics:

- **Turbulence** : Studying and modeling turbulent flows using numerical and experimental methods. The focus is on inhomogeneity and non-stationarity in wall-bounded flows, wakes, jets, and various other flow configurations. This research area includes 5 permanent researchers and around 10 PhD and postdoctoral students.
- **Rotating flows** : Analyzing and modeling internal and external flows related to rotating machines, such as turbo-machinery and rotors. Experimental and numerical studies of rotating machines (turbomachines, rotors, propellers) operating under degraded conditions, as well as flow control, cavitation, and multiphase flows, are also key topics. This subject gathers 6 researchers, 7 research engineers and about 13 PhD and post-doctoral students.
- **Flight dynamics in unsteady and non-uniform environments** : Development of analytical and experimental methods to determine the evolving behaviour of low-speed aircraft in perturbed flight domains. Post-stall dynamics, environmental interactions, and experimental design are among the main research interests. This topic involves about 5 research engineers and 5 PhD and post-doctoral students.

## 1.2 Social and environmental responsibility

ENSAM's Corporate Social Responsibility (CSR) (or Responsabilité Sociétale des Entreprises (RSE)) action plan is structured around five strategic axes[1][2]:

1. **Strategy, governance and territorial anchoring**: presents ENSAM's actions in terms of sustainable development, in particular the Evolutive Learning Factories 4.0 project aimed at modernizing training and research platforms by fully integrating the CSR dimension.
2. **Education and training**: Training students in eco-responsibility
3. **Research and innovation**: Ensure that all research themes have direct or indirect impacts on the achievement of international and national sustainable development goals. Ensure free access to research results and data.
4. **Environmental management**: ENSAM aims to reduce its environmental impact by improving energy efficiency, cutting emissions (including Scope 3), and integrating carbon accounting. It also promotes biodiversity and sustainable mobility through staff engagement, responsible purchasing, and green-space management.
5. **Social policy**: Reconciling economic, societal, and environmental dimensions by training staff, teachers, research professors, and stakeholders in sustainable development issues, developing a quality of life policy within the institution, and ensuring equal opportunities in training, research, and human resources. An indicator is the QVT, that is detailed in the Rapport Social Unique (RSU)[3] 2024.

The "Comité Social et Économique (CSE)" is the staff representative body within ENSAM. Although specific information on the CSE is limited to the 2022 RSU mentioning the monitoring of the actions of the "Comité d'Hygiène, de Sécurité et des Conditions de Travail National (CHSCTN)" which is the old name of the CSE.

Arts et Métiers also publishes its activity reports, its social balance sheet / single social report and public data then transmitted to the Commission des Titres d'Ingénieur (CTI) available online

[1] https://artsetmetiers.fr/fr/ecole-engagee
[2] https://dp-www.s3.ensam.eu/public/2024-11/24_032_plaquette-bilan-RSE_V9_0.pdf
[3] https://dp-www.s3.ensam.eu/public/2025-07/RSU%202024_VCA.pdf

[4]. More specifically, students from Arts et Métiers created a webpage to learn about general CSR and sustainable development: [5] (addressing themes such as ecology, inclusion, gender equality and the fight against discrimination,...).

About the LMFL we can also find an interesting documents from Haut Conseil de l'Évaluation de la Recherche et de l'Enseignement Supérieur (HCERES). This organization is in charge of evaluating all research structures. Every five years, it publishes a report on the LMFL. The last one[6] was published in February of this year. It presents the laboratory's strengths and weaknesses, as well as its research activities, collaborations, workforce, and future perspectives. It highlights the laboratory's commitment to excellence in research, its multidisciplinary approach, and its contributions to the field of fluid mechanics and related areas. More precisely, the report emphasizes the laboratory's strengths in experimental and numerical methods, its expertise in turbulence and rotating flows, and its contributions to the understanding of complex fluid phenomena.

The report highlights that:

- The facilities are unique, with wind tunnels and turbomachinery test rigs that allow advanced experimental investigations of external and internal flows.
- It is one of the most active European laboratories in the field of next-generation aircraft design, particularly on electric propulsion systems, supported by high-level experimental setups such as the Eiffel wind tunnel (L1) and a vertical wind tunnel with a rotating balance. The LMFL also leads European research on airship aerodynamics and urban drone flight.
- Its experimental platforms are completed by advanced numerical simulations, with equipment partly acquired through regional Contrat de Plan État-Région (CPER) programs. The lab conducts original research on turbulence dissipation mechanisms in non-homogeneous, unsteady flows and on turbulent/non-turbulent interfaces.
- Efforts have been made to create synergy between the lab's three geographically separated sites through transversal research themes in optical metrology, flow control, and data analysis. Weekly LMFL webinars support internal collaboration.
- Despite some staff departures, LMFL has maintained its attractiveness by recruiting top researchers, strengthening historical themes like turbulence and instabilities, and expanding into areas like two-phase aerodynamics and data-driven modeling.
- LMFL's scientific output is of high quality, with publications in top journals (e.g., Journal of Fluid Mechanics (JFM)) and participation in major conferences. While flight dynamics publications are fewer, this area is expected to grow through collaborations, particularly with the turbulence team.
- The lab also benefits from significant external funding via competitive calls (ERC, ANR, Clean Sky) and strong industrial partnerships (e.g., CNES, Ariane Group). The unit's leadership has been commended for its organizational work and encouraged to keep fostering synergy across research themes.

[4] https://artsetmetiers.fr/fr/observatoire-des-donnees
[5] https://artsetmetiers.fr/fr/actualites/developpement-durable-et-rse-faites-le-plein-de-connaissances-sur-savoir
[6] https://www.hceres.fr/sites/default/files/media/publications/rapports_evaluations/pdf/E2026-EV-0590349J-DER-ER-DER-PUR260025134-ST5-LMFL-RF.pdf

My personal experience of the work environment has been very positive. The working atmosphere within the laboratory is pleasant and conducive to research. The researchers are passionate about their work and share their knowledge with enthusiasm. The team is close-knit, which encourages the exchange of ideas and collaboration between members from all over the world. The working atmosphere suits me: researchers generally work hard, but there is no pressure regarding working hours. The building is a beautiful old, typical Lille building, but well renovated and adapted to the research work being conducted (both numerical and experimental). The laboratory offers spacious areas, interior courtyards, large lecture halls, and a large workshop for experiments.

# Chapter 2

# Optimizing the performance of a centrifugal pump by data assimilation

When scientistes want to understand the fluide behavior, they can capture physical data by measurements. They can also use physical rules. In very simple cases, analitical solution can be fond. For bit more complex cases, they are able to discretised physical rules to run a simulation. Today, thanks to numerical computation, we are able to simulate the behavior of a fluide. But the computational cost of an "exact" numerical fluide simulation is still too expansive to be realised on industrial cases. This lead researchers to develppe models and technics. Those models are usually composed of varibles that need to be adjute to fit the real world, by doing experiences to capture physical data. One of this technic is the IBM, that will be explane later. This technic use parameters that need to be fetted. We can do this empirically, by trial and error and intuition. This can work for a single parameter. But it time consuming, unprecise and inconceivable for more than few parameters. It existe some mathematical and numerical method to do that, as the EnKF. This lead my supervisors to creat my internship subject.

The internship began in February 2025. I was provided with a fix computer, credentials to connect to the Irene and Cassiopée supercomputers, and the link to the CONES's gitlab[1]. Marcello MELDI, Antoine DAZIN and Miguel MARTINEZ VALERO which are my internship supervisors, showed me around the lab. Then, Tom MOUSSIE and Miguel, my co-officies, explained to me the basics of OpenFoam and CONES, the DA code. I was able to run a few tests on OpenFoam on the supercomputers, which allowed me to familiarize myself with the tools and the code.

J'aimerais ajouter une introduction "pourquoi ce sujet de stage existe ?" (où l'ajouter ?)

## 2.1 Data Assimilation

DA[2] is a technique used to improve the accuracy of numerical models by combining them with observational data. It is widely used in various fields, including meteorology, oceanography, and engineering. The goal of data assimilation is to obtain a more accurate representation of the state of a system by integrating information from different physical measurements or hight fidelity sources.

### 2.1.1 State of the art

It does exist many methods for data assimilation, distributed in three main categories: **variational methods**, **statistical methods** and **hybrid** (see Figure 2.1). Variational methods (also called

[1] https://gitlab.ensam.eu/pe431/cones-dev

classical methods), such as the 3D-Var and 4D-Var, use optimization techniques to find the best estimate of the state of a system by minimizing the difference between model predictions and observations. Statistical methods (also called sequential methods), such as the Kalman filter and its variants (e.g., EnKF), update the model state iteratively as new observations become available. And Hybrid methods that we are not going to explaine here.

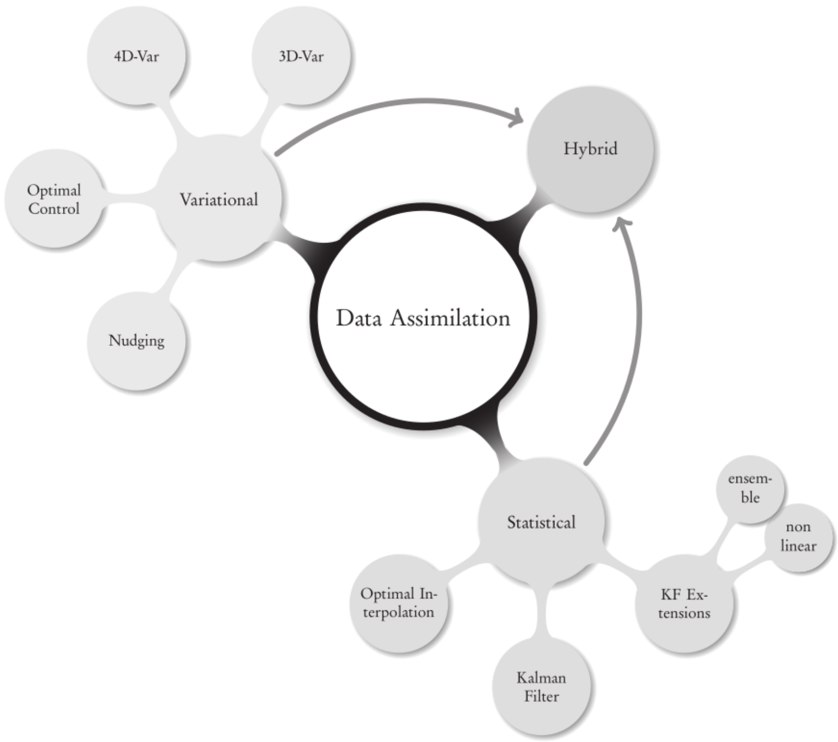



Figure 2.1: The big picture for DA methods and algorithms[2]

**EnKF** Our approach to data assimilation is based on the Ensemble Kalman Filter (EnKF), which is a Monte Carlo method that uses a set of model states (the ensemble) to estimate the state of the system. The EnKF is particularly well-suited for nonlinear systems and has been successfully applied in various fields, including computational fluid dynamics (CFD).

EnKF for DA in CFD has been used at LMFL since a few years, with the work of Lucas Villanueva[10], Miguel M.Valero[10], Tom Moussie[5], Sarp Er, Paolo Errante[5] and Marcello Meldi[10][5] and their colleagues. The EnKF method is a powerful tool for assimilating data into computational fluid dynamics (CFD) simulations, allowing for the correction of model predictions based on observed data. An illustration of the EnKF data assimilation process is shown in Figure 2.2. The maths behind the EnKF method is detailed in the subsection 2.2.3

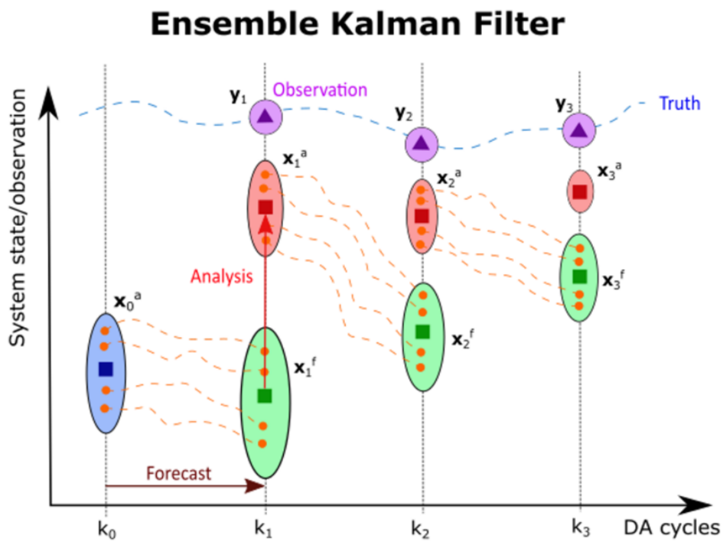
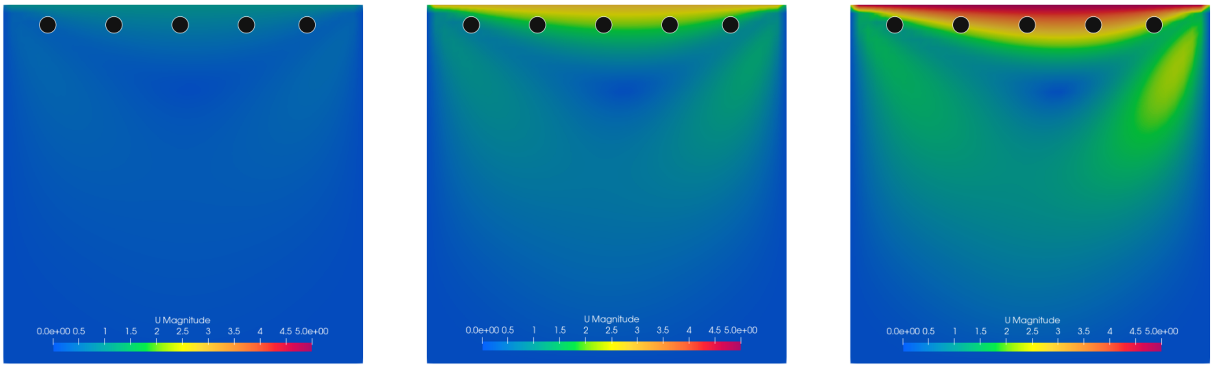



Figure 2.2: Illustration of the EnKF process (100% confidence in observations)

**CFD** Today, we can theorically run a Direct Numerical Simulation (DNS) which resolves exactly the Navier-Stokes equations, but it is still too expensive to be used in practice. Therefore, we generally use Large Eddy Simulation (LES) or Reynolds-Averaged Navier–Stokes equations (RANS) models to simulate the flow. The EnKF method can be applied to these models to adjust specific model parameters or flow fields, enhancing the overall physical consistency of the simulation.

A good example based on the simple and academic *cavity test case* illustrates it well. The cavity test case is a simple flow in a square cavity, where the flow is driven by a moving lid. The flow is characterized by a recirculation zone and a vortex in the center of the cavity. The EnKF method can be used to assimilate data from the flow, such as velocity or pressure measurements, to improve the accuracy of the simulation. The EnKF method can also be used to estimate the parameters of the model, such as viscosity or turbulence model coefficients, to improve the physical accuracy of the simulation.

In our case, we are going to optimise the velocity wall condition. To do so, we use the result on 5 locations of a precise converged simulation with a to boundary condition of 5 m.s$^{-1}$ as a reference. We will then assimilate data from the flow, at the 5 observation points (see Figure 2.3a). We start from a simulation with a wall condition of 1 m.s$^{-1}$, which is not the correct value. The EnKF method will then be used to correct the wall condition by assimilating the data from the flow at the 5 observation points. The EnKF method will iteratively update the wall condition until it converges to the correct value, which is 5 m.s$^{-1}$ in this case. In only 3 EnKF iterations, the wall condition converges to the correct value (with a relative error of 0.05 m.s$^{-1}$).




(a) Initial state (b) After one EnKF it. (c) Converged after 2 EnKF it.

Figure 2.3: Evolution of the state through EnKF iterations (●: observation point)

**IBM** The IBM theory has been developed in 1972 by Charles S. Peskin[6]. The IBM is a numerical technique used to simulate fluid-structure interactions, particularly in cases where the structure is immersed in a fluid flow. This allows to keep numerical precision on complex geometries and moving boundaries without the need for body-fitted meshes, which can be computationally expensive and difficult to implement.

They are different ways to discretize the space when we do CFD (see Figure 2.4[2]). The most common are structured meshes, unstructured meshes and body-fitted meshes. The body-fitted meshes are the most accurate but also the most expensive in terms of computational resources. The IBM allows to use unstructured meshes, which are less accurate but more efficient in terms of computational resources.

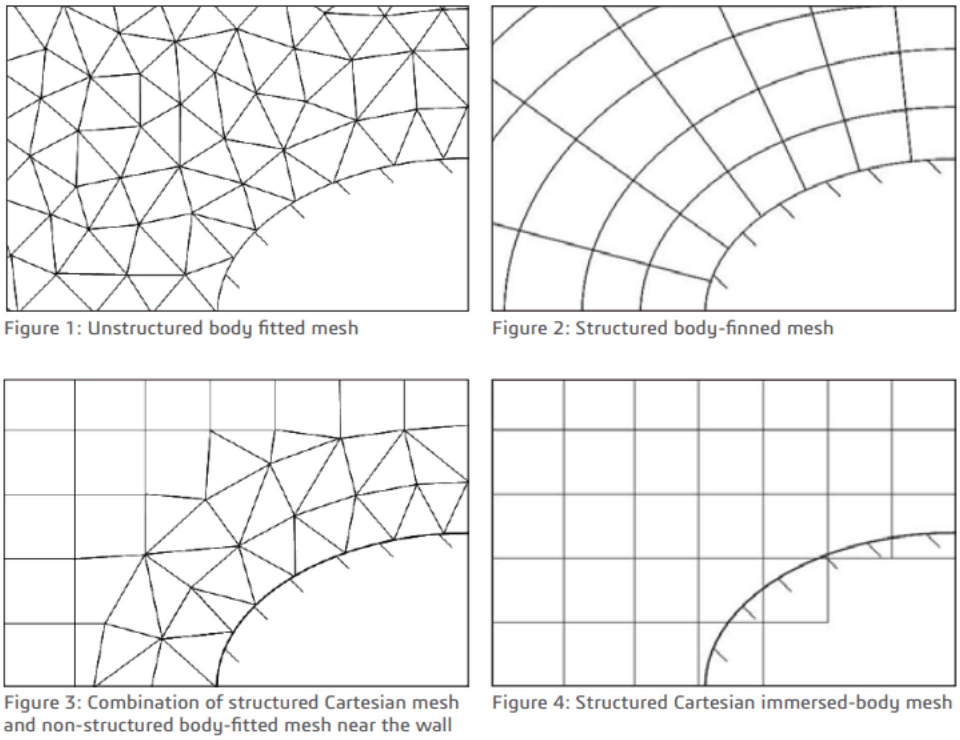



Figure 2.4: Differents ways to mesh

In our case, we impose a forcing term on the Navier-Stokes equations to simulate the effect of the immersed boundary on the fluid flow. This term is add only on the penalization part $\Omega_b U \Sigma_b$

[2] https://www.solidworks.com/sw/docs/flow_basis_of_cad_embedded_cfd_whitepaper.pdf

(see Figure 2.5). The maths behind the EnKF method is detailed in the subsection 2.2.1. The problem of this method is that it leads to a few constants that need to be tuned to obtain the best results. For that, we use the EnKF method to optimize these constants. The EnKF method will be used to assimilate data from the flow at the observation points and to update the forcing term until it converges to the optimal value.

These high fidelity data are obtained from an experiment explained in the next subsection 2.1.2

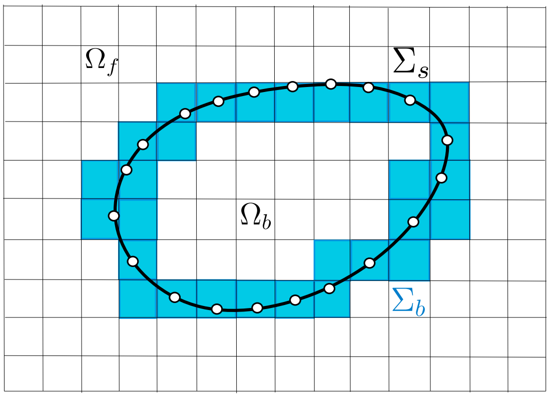


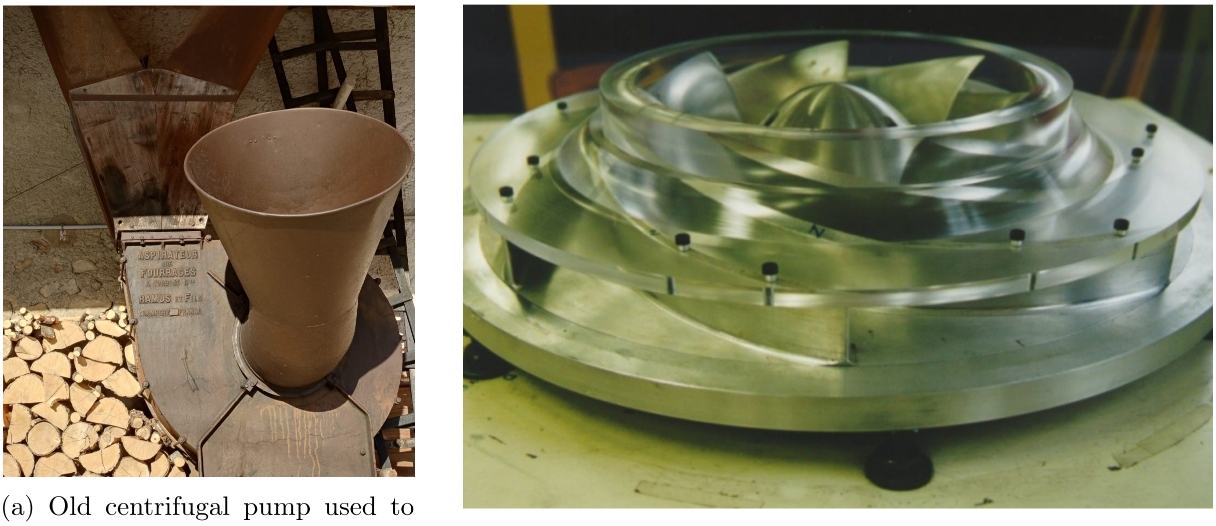

Figure 2.5: IBM regions

This year, my supervisors M. VELRO and M. Meldi published an article[8] about the optimisation of the IBM forcing and turbulence terms, that gives very good results. More recently, they obtained very good results on the optimisation of these parameters on a oscillating cylinder. // Tt ça à vérifier en relisant la thèse de Miguel. //

This led us to the idea of using the EnKF method to optimize the parameters of the IBM forcing term on a centrifugal pump, which is a more complex case than the previous ones. The goal is to improve the performance of the pump by optimizing the parameters of the IBM forcing term, which will lead to a better simulation of the fluid flow and a better understanding of the pump's behaviour.

### 2.1.2 Presentation of the data from the centrifugal pump test case

**The pump** Centrifugal pumps are used since 1475[7] and are still widely used today in various applications, such as water supply, irrigation, turbomachines, and industrial processes. They are known for their low cost, efficiency and reliability, making them a popular choice for many fluid transport applications. The studied centrifugal pump is a Société Hydrotechnique de France (SHF) impeller, which is a type of centrifugal pump designed for high flow rates and low head applications. The RESEDA benchmark is used to study rotor/stator interactions and instabilities in centrifugal machines.



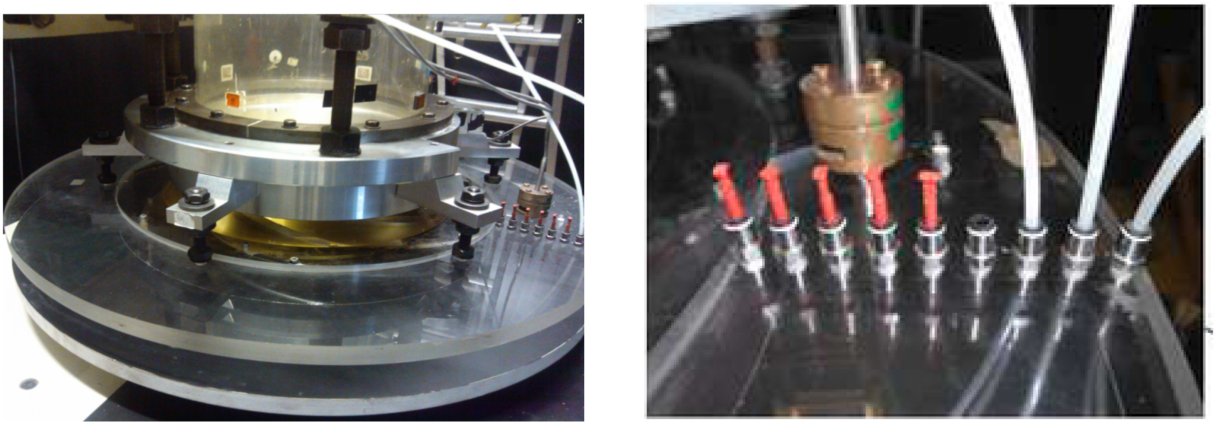
(a) Old centrifugal pump used to move hay (personal picture)

(b) SHF impeller used in the RESEDA benchmark

Figure 2.6: Centrifugal pumps

Different high fidelity data has been recorded on the diffuser of the pump at the LMFL:

- High-Speed Particle Image Velocimetry (PIV) data obtained by Antoine Dazin [3] in 2011;
- Pressure data obtained by Meng Fan [4] in 2022.



(a) Vaneless diffuser of the centrifugal pump

(b) Pressure taps on the diffuser

Figure 2.7: Diffuser and pressure taps of the centrifugal pump

The operating conditions of the machine were :

| | **Impeller characteristics** | |
|---|---|---|
| $R_1$ | Tip inlet radius | $141.1\,mm$ |
| $R_2$ | Outlet radius | $257.5\,mm$ |
| $Z$ | Number of blades | 7 |
| $Q_d$ | Design flow rate | $0.236\,m^3/s$ |
| $K$ | Mean blade thickness | $9\,mm$ |
| $\Omega$ | Rotating velocity | $1\,200\,rpm$ |
| $\beta_2$ | Outside blade angle (from peripheral velocity $U = R_2\Omega$) | $22.5°$ |
| | **Vaneless diffuser characteristics** | |
| $b_3$ | Diffuser width | $38.5\,mm$ |
| $R_3$ | Inlet radius (without leakage, i.e., $R_3 = R_2$) | $257.5\,mm$ |
| $R_4$ | Diffuser outlet radius | $385.5\,mm$ |
| $t$ | Thickness of the upper and lower sections of the diffuser | $20\,mm$ |

Table 2.1: Summary of the main geometrical and operating parameters of the impeller and diffuser at design conditions, employed in the numerical modelling of the centrifugal pump.

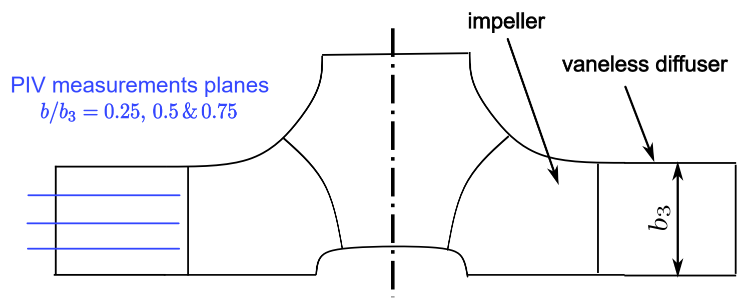



Figure 2.8: Position of the PIV plans on the diffuser

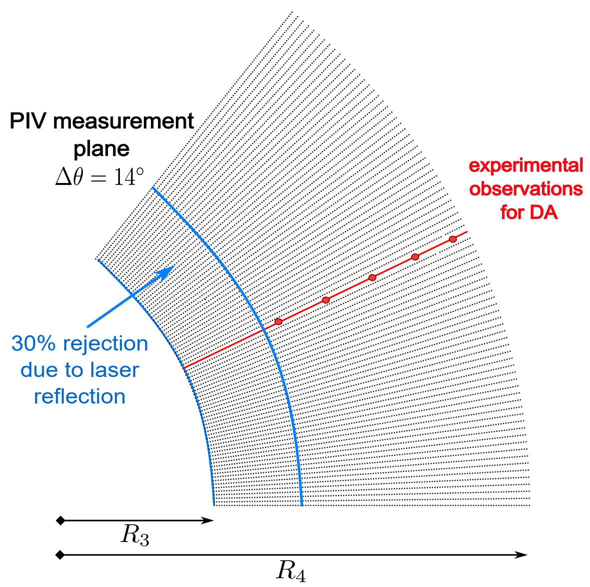



Figure 2.9: Position of the PIV points on a polar slice of the diffuser

**PIV data** Data is stored in a directory structure that contains six directories. The six directories correspond to two tests realised at 25%, 50% et 75% of the diffuser height. Each test was repeated twice for the same operating condition and the same height (R1, R2) In each directory, there are

1581 files corresponding to instantaneous ASCII data file. The ten columns are corrresponding to :

$$x(m), y(m), V_x(m.s^{-1}), V_y(m.s^{-1}), r(m), \theta(rad), Vr(m.s^{-1}), V_\theta(m.s^{-1}), V_z(m.s^{-1}), Unknown$$

And the 10,125 lines correspond to the 81 locations over x and 125 locations over y of the PIV. Processing of these data is detailed in the subsection 2.4.1.

**Pressure data** In addition to the nine pressure tapes on the diffuser, M. Fan has also captured (Figure 2.10) the inlet and outlet pressure that allows to compute the global performance of the pump (see //partie L.N.M à référencer//)

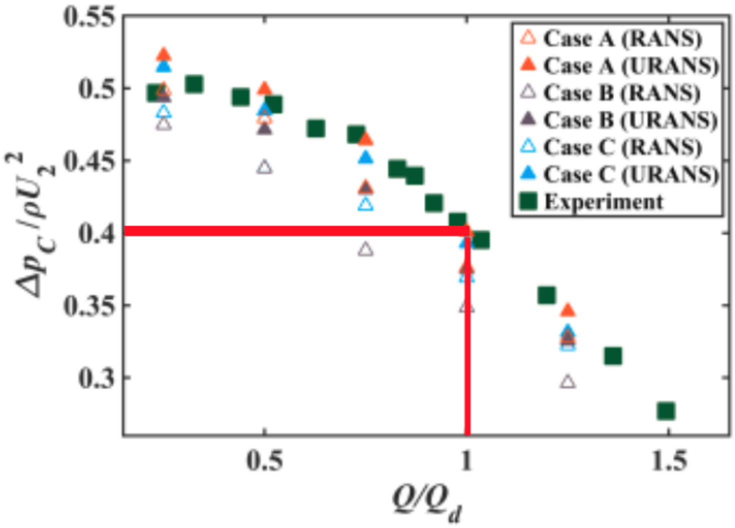



Figure 2.10: Overall pump performance versus reduced pump flow rate Q

As we are at $Q/Q_d = 1$, we can retreive

$$\frac{\Delta P}{\rho U^2} = 0.4 \Rightarrow \Delta P = 0.4 \times \rho \times U^2 = 0.4 \times 1.225 \times 32.24^2 \approx 415 \text{ Pa}$$

## 2.2 Numerical ingredients

### 2.2.1 CFD governing equations

**Navier-Stokes equations** CFD is based on the fundamental equations that describe Newtonian fluid mechanics, base on the the nonlinear Navier-Stokes partial differential equations.

These equations express three physical conservation principles, given here in their Eulerian form (i.e., fixed spatial reference frame), which is the standard in CFD because it describes how fluid properties evolve at fixed locations in space, that is ideal for mesh-based numerical simulations.

- **Mass conservation (Continuity equation):**

$$\frac{\partial \rho}{\partial t} + div(\rho \mathbf{U}) = 0 \tag{2.1}$$

- **Momentum conservation (second Newton Law):**

$$\frac{\partial (\rho \mathbf{U})}{\partial t} + \nabla \cdot (\rho \mathbf{U} \otimes \mathbf{U}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \mathbf{f} \tag{2.2}$$

  where $\boldsymbol{\tau}$ is the viscous stress tensor and $\mathbf{f}$ includes body forces (e.g., gravity, source terms).

- **Energy conservation (First law of thermodynamics):**

$$\frac{\partial (\rho E)}{\partial t} + \nabla \cdot (\mathbf{U}(\rho E + p)) = \nabla \cdot (k \nabla T) + \Phi \tag{2.3}$$

  where $E = e + \frac{1}{2}\mathbf{U}^2$ is the total energy per unit mass, $k$ is the thermal conductivity, $T$ the temperature, and $\Phi$ the viscous dissipation.

As we considere that we are in **incompressible** flow (because we are at low mach numeber), Equation 2.1 becomes

$$\nabla \cdot \mathbf{u} = 0 \tag{2.4}$$

**Turbulence modeling** This equations permit to run a DNS simulation, that would be too costly in our case. Instead of running simulation that precision go down to the Kolmogorove scale, we use classical RANS CFD modelisation. The Figure 2.11[3] and Figure 2.12[4][5] illustrates well the tree main differents type of CFD models.

[3] https://www.cfdsupport.com/openfoam-training-by-cfd-support/node346/
[4] https://blog.diphyx.com/comprehensive-guide-review-to-choosing-the-right-cfd-software-in-2023-features-performan
[5] https://help.altair.com/hwcfdsolvers/acusolve/topics/acusolve/training_manual/turbulence_modeling_r.htm

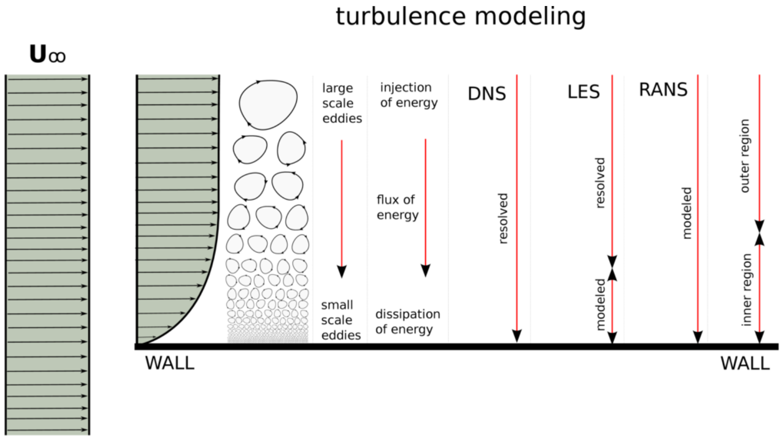
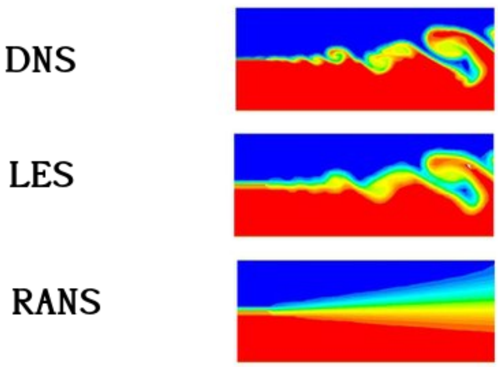
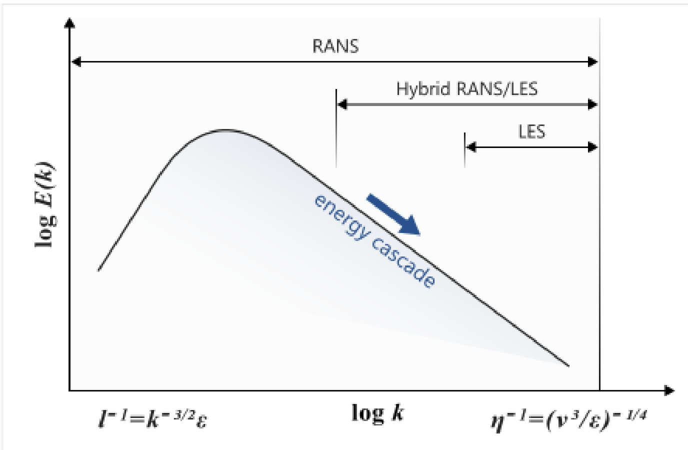



Figure 2.11: Modelisation of turbulence




(a) Illustration of different turbulence models




(b) Turbulence model on the energy cascade

Figure 2.12: Turbulence models

Kolmogorove Scale

In our simulation, we are going to use a $k-\omega$ Shear Stress Transport (SST) turbulence model. Is a two-equations eddy-viscosity model: it's combine the $K-\epsilon$ and the $K-\Omega$ turbulence approximations.

$K-\Omega$ is know to be a good model in near-wall region but a bad prediction in bulk flow. $K-\epsilon$ make a good numerical prediction in the bulk flow field but is less accurate near solid surfaces

Kinematic Eddy Viscosity[6] :

$$\nu_T = \frac{a_1 k}{\max(a_1\omega, SF_2)}$$

Turbulence Kinetic Energy :

$$\frac{\partial k}{\partial t} + U_j\frac{\partial k}{\partial x_j} = P_k - \beta^* k\omega + \frac{\partial}{\partial x_j}\left[(\nu + \sigma_k\nu_T)\frac{\partial k}{\partial x_j}\right]$$

Specific Dissipation Rate :

$$\frac{\partial \omega}{\partial t} + U_j\frac{\partial \omega}{\partial x_j} = \alpha S^2 - \beta\omega^2 + \frac{\partial}{\partial x_j}\left[(\nu + \sigma_\omega\nu_T)\frac{\partial \omega}{\partial x_j}\right] + 2(1-F_1)\sigma_{\omega 2}\frac{1}{\omega}\frac{\partial k}{\partial x_i}\frac{\partial \omega}{\partial x_i}$$

[6] https://www.cfd-online.com/Wiki/SST_k-omega_model

Closure Coefficients and Auxilary Relations :

$$F_2 = \tanh\left[\left[\max\left(\frac{2\sqrt{k}}{\beta^*\omega y}, \frac{500\nu}{y^2\omega}\right)\right]^2\right]$$

$$P_k = \min\left(\tau_{ij}\frac{\partial U_i}{\partial x_j}, 10\beta^* k\omega\right)$$

$$F_1 = \tanh\left\{\left\{\min\left[\max\left(\frac{\sqrt{k}}{\beta^*\omega y}, \frac{500\nu}{y^2\omega}\right), \frac{4\sigma_{\omega 2}k}{CD_{k\omega}y^2}\right]\right\}^4\right\}$$

$$CD_{k\omega} = \max\left(2\rho\sigma_{\omega 2}\frac{1}{\omega}\frac{\partial k}{\partial x_i}\frac{\partial \omega}{\partial x_i}, 10^{-10}\right)$$

$\phi = \phi_1 F_1 + \phi_2(1 - F_1)$

$\alpha_1 = \frac{5}{9}, \quad \alpha_2 = 0.44$

$\beta_1 = \frac{3}{40}, \quad \beta_2 = 0.0828$

$\beta^* = \frac{9}{100}$

$\sigma_{k1} = 0.85, \sigma_{k2} = 1$

$\sigma_{\omega 1} = 0.5, \sigma_{\omega 2} = 0.856$

**MRF method** To simulate the rotation of the pump, with stationary RANS result, we use a Multiple Reference Frame (MRF) method.

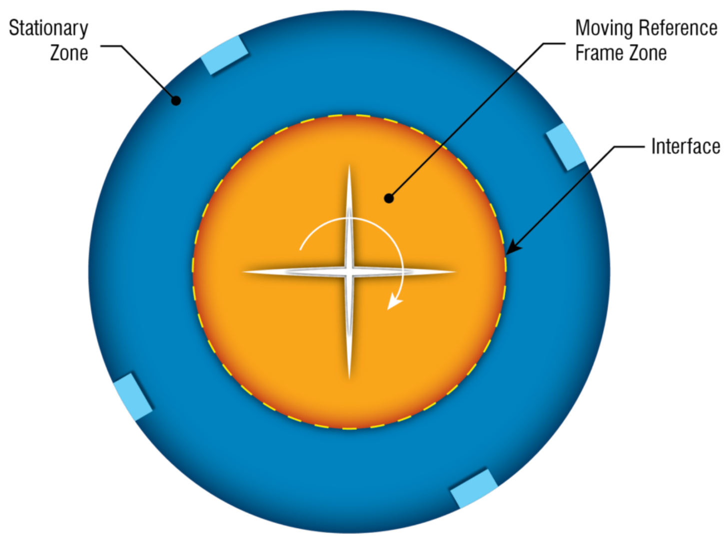



Figure 2.13: Turbulence

The principle is to add relative frame terme in a moving part while the other part stay fixe. Neither the mesh or the body is moving.

In the U-RANS case, the impeller is moving. Also, very large eddy are simulated.

To do so, the CFL hac Courant-Friedrichs-Lewy (CFL) or the courant number has to be computed. Typically set to 0.5 for stability. A way to visualize this is to think of it as the fraction of the distance a wave travels in one time step relative to the distance between two cells in the azimuthal direction at the tip of the impeller.

- $\Delta_x$ is the distance between two cells in the azimuthal direction at the tip of the impeller in our model, which is given as 1.5 mm or 0.0015 m.

- $\|U_\infty\|$ is the magnitude of the velocity at infinity, which can be approximated by the tangential velocity at the tip of the impeller.

$$\|U_\infty\| = \omega \cdot r_{tip} \tag{2.5}$$

where:

- $\omega$ is the angular velocity in radians per second,

- $r$ is the radius at the tip of the impeller.

Assuming the rotating speed is given in revolutions per minute (rpm), we can convert it to radians per second:

$$\omega = \frac{1200 \text{ rpm} \cdot 2\pi \text{ rad}}{60 \text{ s}} = 125.66 \text{ rps.} \tag{2.6}$$

Assuming the radius at the tip of the impeller is $r = 0.2566$ m (which is 0.2566m), we can compute the tangential velocity:

$$\|U_\infty\| = 125.66 \text{ rps} \cdot 0.2566 \text{ m} = 32.24 \text{ m/s.} \tag{2.7}$$

Assuming the distance between two cells in the azimuthal direction at the tip of the impeller is $\Delta_x = 0.0015$ m (which is 1.5 mm), we can now compute the time step $\Delta_t$:

$$\Delta_t = \frac{0.5 \cdot 0.0015 \text{ m}}{32.24 \text{ m/s}} = 2.3 \times 10^{-5} \text{ s} \tag{2.8}$$

Annexe des opérateurs grad, div, nabla, rot, laplacien, laplacien vectoriel, etc... Mentionner que les solutions numériques ne sont pas démontrés

**IBM equations** As see in the section subsection 2.1.1, instead of use bodyfitted mesh, we are going to consider porous medium and adding a forcing terme in the physical part (see Figure 2.5).

$$(\boldsymbol{u} \cdot \boldsymbol{\nabla})\, \boldsymbol{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \boldsymbol{u} \underbrace{-\xi \nu \boldsymbol{D}\,(\boldsymbol{u} - \boldsymbol{u}_{ib})}_{\boldsymbol{f}_P} \tag{2.9}$$

Here, we are dividing both side of the eqautions by $\rho$ for simplicity as we are in incompressible flow. So $[p] = M \cdot L^{-1} \cdot T^{-2} and [\boldsymbol{u}] = L \cdot T^{-1}$

$$\xi = \begin{cases} 0 & \text{if } \boldsymbol{x} \in \Omega_f \\ 1 & \text{if } \boldsymbol{x} \in (\Sigma_b \cup \Omega_b) \end{cases} \tag{2.10}$$

To fit the mathematical solution, $D$ have to tend to the infinite. But this lead to numerical instabilities.

Methodo (forcage continue ici (vs interpolation ?)) -> avantages

Darcy's law in the body interface region $\Sigma_b$ and the solid region $\Omega_b$. To do so, a spatial dependent tensor $\boldsymbol{D}(\boldsymbol{x})$ must be defined. The resulting penalty term is modelled as follows:

$$\boldsymbol{f}_P = \begin{cases} \boldsymbol{0} & \text{if } \boldsymbol{x} \in \Omega_f \\ -\nu\, \boldsymbol{D}\, (\boldsymbol{u} - \boldsymbol{u_{ib}}) & \text{if } \boldsymbol{x} \in (\Sigma_b \cup \Omega_b) \end{cases} \tag{2.11}$$

$\boldsymbol{u_{ib}}(\boldsymbol{x}, t)$ is a target velocity representing the immersed body's physical behaviour. ... $\boldsymbol{u_{ib}} = \boldsymbol{0}$. The components $D_{ij}$ of the tensor $\boldsymbol{D}$ are usually controlled to obtain the best compromise between the accuracy and stability of the numerical solver

$$Re_\Omega = \frac{R_2^2 \Omega}{\nu} = 5.53 \times 10^5$$

$$Ma = \frac{R_2 \Omega}{c} = 0.095 \ \textbf{(incompressible)}$$

Reprendre ma présentation (dimensiosn de p)

$$\boldsymbol{\nabla} \cdot (\boldsymbol{u}\boldsymbol{u}) = -\nabla p + \nu \nabla^2 \boldsymbol{u} \overbrace{- \frac{\xi}{\eta} (\boldsymbol{u} - \boldsymbol{u_{ib}})}^{\boldsymbol{f}_P}$$

$$\xi = \begin{cases} 0 & \text{if } \boldsymbol{x} \in \Omega_f \\ 1 & \text{if } \boldsymbol{x} \in (\Sigma_b \cup \Omega_b) \end{cases}$$

### 2.2.2 The open source code OpenFoam

Simulations are performed using the C++ framework OpenFOAM, used to discretize the equations previously presented. OpenFOAM is an open-source which provides many tools, referred to as executables

- **Pre-processing**, including meshing and decomposing tools.
- **Solving**, including in-built and custom solvers. Each solver is built to answer a specific CFD problem.
- **Post-processing**, including `ParaView`, an open-source visualization application.

OpenFOAM is not provided with a graphical interface, which is why a simulation is built by writing "dictionaries", in which are stored the necessary information for the simulation (mesh, turbulence model, maximum number of characteristic times, ...).

See Annex 2.

Ou si plus lisible :

```
<case>
├── 0
│   ├── k
│   ├── omega
│   ├── p
│   └── U
├── 0.orig
│   └── 0
├── constant
├── system
│   ├── blockMeshDict
│   ├── controlDict
│   ├── decomposeParDict
│   ├── fvSchemes
│   └── fvSolution
├── tests
└── <time directories>
```

Figure 2.14: Arbo of OpenFoam

The `system` directory contains the main parameters concerning the progress of the simulation:

- The `controlDict` dictionary defines parameters such as start and end time, time steps, data output formatting, or data average computation.
- The `blockMeshDict` dictionary defines the geometry, mesh and boundary conditions implementation.
- The `decomposeParDict` dictionary defines the geometry and mesh decomposition, with the objective to run the simulation in parallel, using multiple cores.
- The `fvSchemes` dictionary defines the discretisation schemes used in the solution.
- The `fvSolution` dictionary defines the equation solvers, tolerances and other algorithm controls for instance.

The `constant` directory contains the different physical properties of the simulation. Moreover, the subfolder `polyMesh` contains the mesh data, obtained after the `blockMesh` command line is run:

- The `momentumTransport` dictionary defines the simulation type (RANS or LES), as well as the turbulence model.
- The `transportProperties` dictionary defines the transport model (Newtonian for instance) and some transport coefficients, such as the kinematic viscosity $\nu$.
- The `MRFProperties` dictionary defines in our case to specifie the rotating zone with the origin, rotation axis and rotation speed.

The `0` directory contains the initial solution's fields. A compiled solver can be call in the terminal on the case directory to run the simulation. If it's called, the `postProcessing` directory stores data from the running simulation, such as forces and probes' data. The `processor0...N-1` directories directly result from the geometry decomposition, which is proceeded through the `decomposeParDict` dictionary. The `dummy.foam` is an empty file which allows to visualize the simulation with `ParaView`.

To implemente the IBM forcage, we use the open source code OpenFoam, which is a C++ library for CFD. OpenFoam provides a wide range of tools for solving the Navier-Stokes equations, including solvers, meshing tools, and post-processing tools. It is widely used in the CFD community and has a large user base. It's developped since 2004 and is composed today of more than a million lines.

More precisely, for example, to implemente the IBM forcage on the speed U, we add the forcing term $\boldsymbol{f}_P$ as follows:

```
// Momentum predictor step

MRF.correctBoundaryVelocity(U) // Corrects U at boundaries to include MRF effects
                               // (e.g., rotating walls)

tmp<fvVectorMatrix> tUEqn
(
    fvm::div(phi, U)            // Convective term
  + MRF.DDt(U)                  // Time derivative in the rotating reference frame
  + turbulence->ddivDevSigma(U) // Turbulent diffusion term
  + cmptMultiply(D, U - UIB)    // Forcing term: D * (U - UIB)
 ==
    fvModels.source(U)          // Source terms from external models
                                // (e.g., body forces), here equal to 0
);
fvVectorMatrix& UEqn = tUEqn.ref();

UEqn.relax() // Applies under-relaxation for numerical stability

fvConstraints.constrain(UEqn); // Applies physical constraints
                               // (e.g., boundary conditions)
```

With - *cmptMultiply* the forward product - D -> esilon x viscosity x D - U the velocity of the field - UIB the velocity of the walls (here omega R)

To compute the time step $\Delta_t$ of the pimpleFoam simulation of the pump, we can use the CFL condition, which states that:

$$\Delta_t = \frac{CFL \cdot \Delta_x}{\|U_\infty\|}, \tag{2.12}$$

where: $CFL$ is the Courant number, detail in the subsection 2.2.1

### 2.2.3 EnKF

As seen in the subsection 2.1.1, we are going to use the stochstic EnKF to assimilate the high fidelity data.

The estimation is calculated by using a set of observations $\mathbf{y}$, a prior state $\mathbf{x}$, and a linear model $\mathbf{M}$. The estimation state is obtained through two operations, the forecast and the analysis, respectively using the superscript f and a. The first forecast step at time $\mathbf{k}$ computes the estimated state $\mathbf{x}_{k+1}$ and the error covariance matrix $\mathbf{P}_k^f$.

The first forecast step at time $k$ computes the estimated state $\mathbf{x}_{k+1}^f$ and the error covariance matrix $\mathbf{P}_{k+1}^f$,

$$\mathbf{x}_{k+1}^f = \mathbf{M}\mathbf{x}_k \tag{2.13}$$

$$\mathbf{P}_{k+1}^f = \mathbf{M}_{k+1}\mathbf{P}_k^a\mathbf{M}_{k+1}^\top + \mathbf{Q} \tag{2.14}$$

where $\mathbf{Q}$ represents the variance matrix of the model. Then, the analysis step in which the observation $\mathbf{y}$ and forecast are combined,

$$\mathbf{K}_{k+1} = \mathbf{P}_{k+1}^f\mathbf{H}^\top\left(\mathbf{H}\mathbf{P}_{k+1}^f\mathbf{H}^\top + \mathbf{R}_{k+1}\right)^{-1} \tag{2.15}$$

$$\mathbf{x}_{k+1}^a = \mathbf{x}_{k+1}^f + \mathbf{K}_{k+1}\left(\mathbf{y}_{k+1} - \mathbf{H}\mathbf{x}_{k+1}^f\right) \tag{2.16}$$

$$\mathbf{P}^a_{k+1} = (\mathbf{I} - \mathbf{K}_{k+1}\mathbf{H})\mathbf{P}^f_{k+1} \tag{2.17}$$

where $\mathbf{R}$ represents the variance matrix of the observations, and $\mathbf{H}$ a linear operator which maps the true state space into the observed space.

**Algorithm 1** Scheme of the EnKF used in the present analyses for both state estimation and parameter optimisation.

**Input:** $\mathcal{M}$ (model); $\boldsymbol{R}$ (observation covariance); $\boldsymbol{y}_k$ (observations);
$\boldsymbol{u}'_{i,0} = [\boldsymbol{u}_{i,0}, \theta_{i,0}]^T$ (initial prior states with $\boldsymbol{u}_{i,0} \sim \mathcal{N}(\overline{\boldsymbol{u}}_0, \boldsymbol{P}_0)$ and $\theta_{i,0} \sim \mathcal{N}(\overline{\theta}_0, \boldsymbol{P}_0)$)
$k = 1, 2, ..., K$ is the iteration index; $i = 1, 2, ..., N_e$ is the ensemble index
**for** $k = 1$ to $K$ **do**
    **for** $i = 1$ to $N_e$ **do**
        Advance in time: $\boldsymbol{u}^f_{i,k} = \mathcal{M}(\theta_{i,k-1})\,\boldsymbol{u}_{i,k-1}$
        Observation: $\boldsymbol{Y}_{i,k} = \boldsymbol{y}_k + \boldsymbol{e}_i$, $\boldsymbol{e}_i \sim \mathcal{N}(0, \boldsymbol{R})$
        Compute ensemble means: $\overline{\boldsymbol{u}}'^f_k = \frac{1}{N_e}\sum_{i=1}^{N_e} \boldsymbol{u}'^f_{i,k}$, etc.
        Anomaly matrices: $\boldsymbol{X}_k = \frac{\boldsymbol{u}'^f_{i,k} - \overline{\boldsymbol{u}}'^f_k}{\sqrt{N_e - 1}}$, etc.
        Kalman gain: $\boldsymbol{K}_k = \boldsymbol{L} \odot \boldsymbol{X}^f_k (\boldsymbol{S}_k)^T [...]$
        Update: $\boldsymbol{u}'^a_{i,k} = \boldsymbol{u}'^f_{i,k} + \boldsymbol{K}_k(\mathbf{Y}_{i,k} - \mathcal{H}(\boldsymbol{u}^f_{i,k}))$
        Covariance inflation: $\boldsymbol{u}'^a_{i,k} = \overline{\boldsymbol{u}}'^a_k + \lambda_k(\boldsymbol{u}'^a_{i,k} - \overline{\boldsymbol{u}}'^a_k)$
    **end for**
**end for**

Préciser que dcp dans mon cas, que paramètres, et tt les observations même par rapport au temps Préparer l'introduction des variables du controlDict et le CONESDict

### 2.2.4 The CONES library

CONES is a library coded in Python/C++ developed for "Coupling OpenFoam with Numerical EnvironmentS" since 2022 at LMFL. Is the only code that permit online DA in CFD codes (it use Message Passing Interface (MPI) process to do it). It's parallelized and can be used on supercomputers. It is mainly designed to work with OpenFoam and DA codes as EnKF.

Thus, a specific directory structure must be observed, as outlined in ...

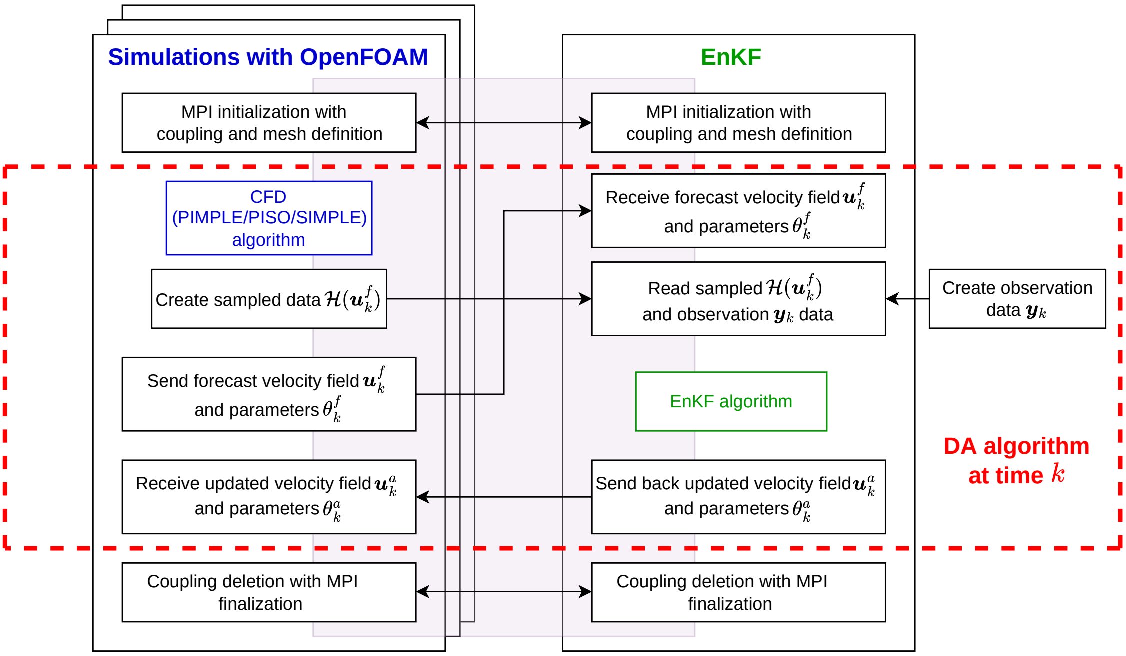



Figure 2.15: CONES structure

### 2.2.5 Pump modeling

The objective of the model is to try to find same physical behavior as the experimental data. We call that digital twin.

For that we have first to discretize the numerical model of the pump. The mesh is generated using the `blockMesh` utility, which is a built-in OpenFOAM tool. The mesh is composed of around 11 million cells. The mesh is composed of hexahedral cells. (quasi only, exepted some interface, cf Figure 2.16). We don't need complex mesh because we dont neek to fit the geometry of the impeller.

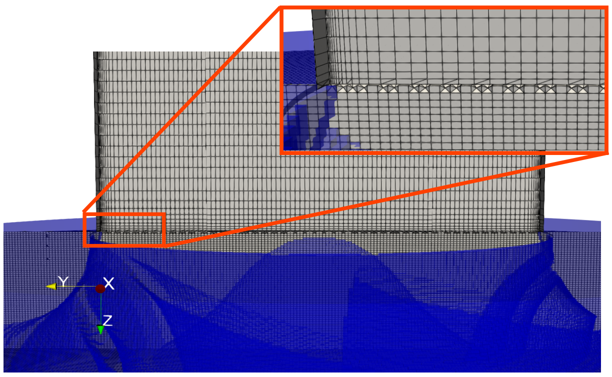



Figure 2.16: Mesh interface between the pipe and the impeller

Boundary conditions are defines as that:

- Inlet: velocity inlet
- Outlet: pressure outlet
- Wall: no-slip wall

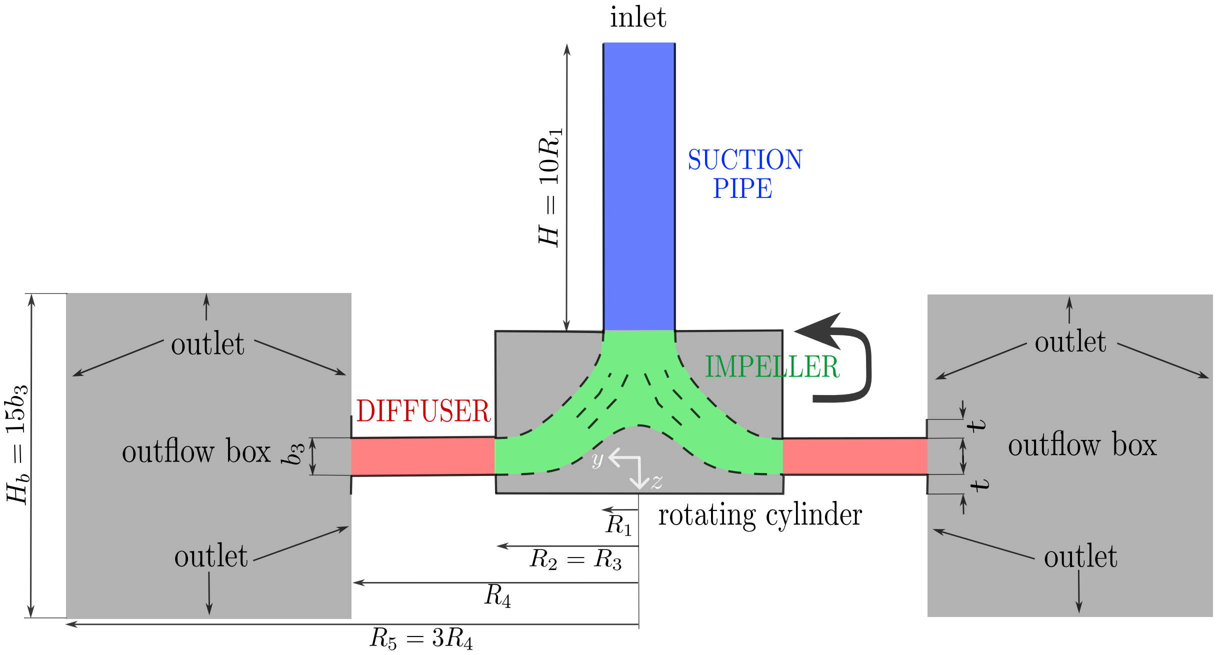



Figure 2.17: Model of the RESDA benchmark

## 2.3 IBM preliminary results

With the conditions specified in the subsection 2.1.2, and a value of 3e7 for the all five forcing parameters, we obtain these velocity and pressure profils.

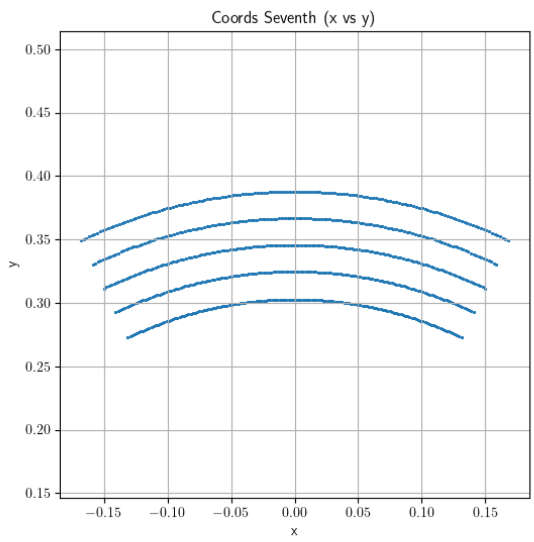



Figure 2.18: Interpolation points over 360/7 degrees

As seen in the subsection 2.1.2, Figure 2.9, we use five observations points on the three PIV plans. We are able to plot this curves, comparing the velocities of the simulation and of the PIV data.

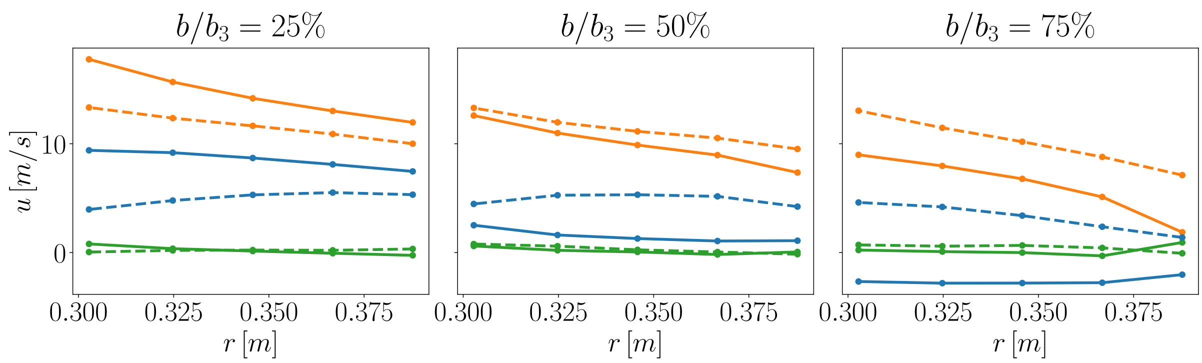



Figure 2.19: Comparison between the profiles inside the diffuser for the numerical simulation (continuous lines), and the PIV data (dashed lines). In particular, (——) represents the radial velocity $u_r$, (——) depicts the azimuthal velocity $u_\theta$, and (——) illustrates the vertical velocity $u_z$.

Our goal is to make these curves fit as well as possible. We already see that there is a notable difference bteween the radial velocity in the plan at 75%. The numerical simulation predict a negative velocity instead of a positive one. This is explaned by a non physical recirculation buble as you can see in the

A good indicator of the accuracy of the simulation is the performance of the pump. It's directly linked to the pressure difference between the outlet and the inlet. As seen in at the end of the subsection 2.1.2, we should obtain around 415Pa. In our case, we obtain a pressure difference of around 306Pa, that is far from what we excpect.

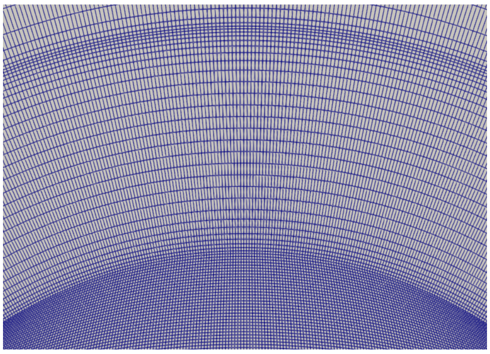


Figure 2.20: diffuseur Mesh

Limitations IBM à améliorer avec DA Graçe aux données xp (mentionné partie 2.1.2, on peut faire de la DA)

## 2.4 Application of the DA

Explication de coords_seventh.txt, coords_360.txt et coords_7.txt

In this work, we aim to improve the numerical model of a centrifugal pump using experimental data obtained from PIV. On one side, we have a numerical simulation of the pump (the digital twin), and on the other, experimental PIV measurements. To combine the strengths of both, we use data assimilation, a method that updates a numerical model using real-world data. We specifically use the EnKF, a data assimilation technique that relies on an ensemble of simulations to estimate and reduce uncertainty in the model. The process can be summarized in two main steps:

1. **Forecast Step** We run a steady-state numerical simulation of the pump using the Multiple Reference Frame (MRF) and the IBM methods. This yields an ensemble of model states, representing the expected flow field within the pump.
2. **Analysis Step** We compare the simulation results with the experimental PIV data. To do this, we interpolate the PIV measurements onto the mesh of the numerical model, creating an ensemble of observation vectors. We then compute the difference between the simulated and measured data, known as the innovation (see mathematical formulation in subsection 2.2.1). This innovation is used to update the ensemble through the EnKF algorithm, improving the numerical model.

1. **Nature of the Numerical Simulation** The simulation is stationary (time-independent) and uses the MRF approach to model rotation. While the velocity field remains constant over time, it varies in space—especially in the diffuser region. Because of the pump's geometry, we expect a periodic velocity pattern that repeats every 1/7th of a revolution (due to the 7 blades). However, this repeating pattern is not clearly visible in unsteady simulations due to wake deformation.
2. **Nature of the Experimental Data** The PIV system captures 49 images per full rotor revolution, meaning there are 7 images between two successive blades. In the rotating frame of reference, each image corresponds to a phase shift of approximately 360°/49.

Ideally, to compare consistent datasets, we could run a time-resolved unsteady simulation and synchronize it with the PIV snapshots. However, for this initial study, we chose to work with a steady MRF simulation.

Although we could attempt to reconstruct the spatial velocity field from the phase-averaged PIV data (as done in [citer article Antoine]), there are two main limitations:

To address these issues, we opted to average the PIV data over time, which is equivalent to spatial (azimuthal) averaging in the rotating frame. On the simulation side, we therefore perform azimuthal averaging of the numerical results to make them consistent with the experimental data.

This approach ensures that the data fed into the EnKF—both the forecast and the observations—are comparable and compatible in both spatial and statistical sense.

Imager avec des illustrations et vue en coupe ParaView

Pour cela j'ai commencé en moyennant 1/7 mais les paramètres étant locaux (non globaux), j'ai ensuite moyenné sur 360 degrés, ce qui permet de mieux prendre en compte les variations locales des paramètres. Cependant, nous pouvons converger vers une moyenne physique qui atteinds un minimum local mais qui ne respecte pas d'avoir 7 fois le même pattern. Cela permet à Fx d'être très différent de Fy mais aussi à la simulation de diverger avec des paramètres et vitesses erronéees. Pour palier ce problème, je compare la moyenne des patternes sur mes 7 segments de 360/7 degrés.

Nous avons fait l'EnKF sans l'état... Sans inflation pour commencer. Si on converge vers une solution qui ne nous satisfait pas, on peut essayer d'ajouter... Cela permet de s'échaper d'un minimum local en ajoutant du bruit artificiellement.

Nous avons fait une simu RANS mais nous aurion pu faire une simu URANS (ou LES ou DDES) avec un modèle de turbulence...

L'expérience montre un ordre de grandeur de centaines d'itération entre les phases d'analyse, mais cela dépend beaucoup du cas. Il en est de même pour le nombre de phases d'analyse. Supposons 1000 itération en moyenne entre chaque phase d'analyse, et 300 phases d'analyse, cela nous fait 300 000 itérations à simuler, soit 93h, sur env. 2800 coeurs, soit 264kh.processeur de calcul.

Comment calculer la variance que je veux mettre ? Expérience : 5% De mon coté : je vise qu'en moyenne un membre d'ensemble ait un écart de 1e7 par rapport au temps précédent (pour des valeurs autour de 5e7), donc, pour 30 membres d'ensemble, environ 95% = 2 sigma dans 5e7+-1e7, càd sigma = 0.5e7 = 10%. Ensuite, le nombre d'itération entre les phases d'analyse est déterminé en regardans combien d'itération sont nécessaires pour faire varier la solution dans le diffuseur (influe, rapidement, converge doucement mais ce qui nous intéresse c'est que ça prenne la tendace) lorsqu'un paramètre de forcage est modifié (supposé très inférieure pour la turbulence car appliquée de partout)

The observation window was defined by observing the number of iterations required for the influence of the IBM to reach the diffuser and take the form of the converged result. We are not interested in the turbulence parameters to determine the observation window because they are present throughout the domain, their influence is almost direct (not necessarily their convergence). We chose 800 iterations

Pb de la DA : plus rapide que DNS mais toujours plus lent que RANS et le temps d'attente et de calcul sur supercalculateur (sur 4000 processeurs) est de l'orde de plusieurs jours. On peut donc se demander si la DA est vraiment utile dans ce cas. En effet, la DA est plus rapide que la DNS mais pas que la RANS. Cependant, il faut prendre en compte le fait que la DA permet de réduire le temps de calcul de la simulation numérique en permettant d'optimiser les paramètres de l'IBM. De plus, la DA peut être utilisée pour d'autres cas d'étude où la DNS est trop coûteuse.

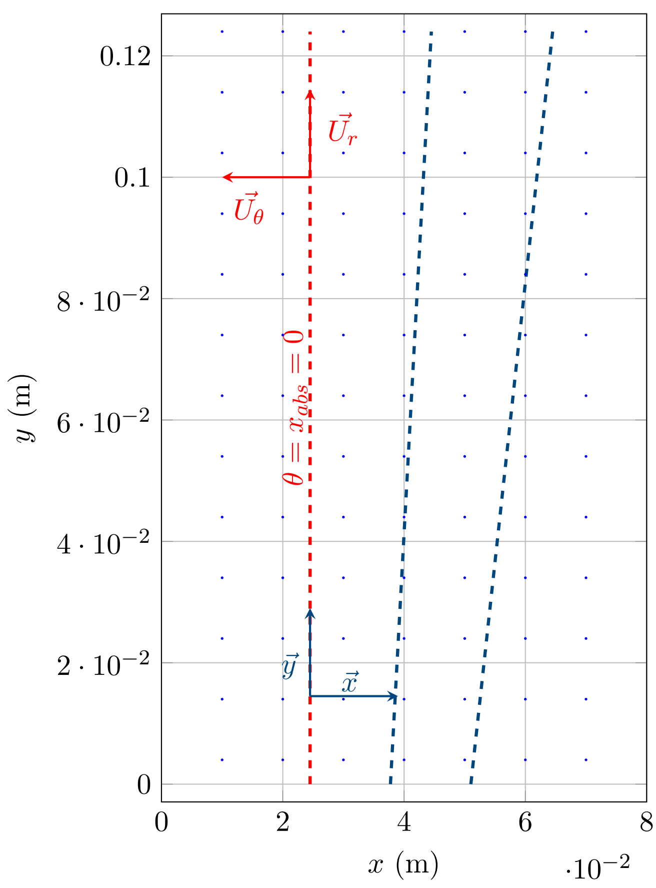
Pb perso : jusqu'à plusieurs jours d'attente pour lancer une simu.

### 2.4.1 Preprocessing of PIV data

#### PIV acquisition treatment

The PIV data is acquired in a rectangular zone of 0.08 m x 0.124 m. The cathesian coordinates are given in a local system, where the absolute origin is at the center of the pump.

Zone d'acquisition PIV




Dans la simulation numérique, l'axe z est vers le centre de la Terre, contrairement à l'expérience où il est vers le haut. La roue tourne dans le sens trigonométrique dans le repère de la simulation numérique et dans le sens antitrigonométrique dans l'expérimentation. Actuellement, je respecte le formalisme de la simu num pour le repère.

Parler des coefficients de relaxation.

**1.** First, we need to know how to convert the coordinates from the local system to the absolute system. We find that between the line 55 and 56, the radius is constant and pass from lower than $\pi$ to upper. In fact, for the plan at 75%, it is the same approache but between the line 65 and 66 because of some technical experimental constraints. We also did plot x over theta for three differents values of theta and find out an intersection point for x = 0.0245. So now, we know that for x = 0.0245, theta = $\frac{\pi}{2}$

$$\begin{cases} x_{rel} = x_{abs} + 0.0245 \\ y_{rel} = x_{abs} + R_2 \end{cases} \tag{2.18}$$

**2.Average** For now we are in stationary case (time invariant). So we are going to deal with data average over time. There are around 49 maps (=file) by turn. So to agerage the data over the time, we can average over the $\phi$ ($\phi$ appartiens à 0-211rd) (=maps). That into the six directory.

**3.Sampling** We will also average data between R1 and R2 to go to 3 files (of 10x81x125 elements). Now, we keep only the data on the line x = 0.0245 (in fact we average data between x

= 0.024 and x = 0.025). Data are then truncate at line 89, where the radius is upper than 0.3, because of some laser reflexion on the pump during the PIV experiment.

**4.Shaping for CONES** Cones keep data in a particular way. We need to convert the data in a format that cones can read. From 3 files (of 10x81x125 elements), to the files `fieldCylindrical.txt` and `obs_coordinates.txt`. Describe shapes...

**5.Creation of a file for the interpolation** Finally, we need to create an `coords_seventh.txt` file for the interpolation function in CONES. We don't juste use the coordinates of the points in the PIV data, beacause we want to average over the angle $\theta$ on the pump. The numerical simulation is $\frac{2\pi}{7}$ OU $2\pi/7$ periodic, so we interpolate in 150 points over theta between 0 and 2pi/7. To do that, wue just take the `obs_coordinates.txt` file and duplicate 150 times the points over the angle $\theta$ (between -pi/7 and pi/7). In our case, we chose 150 that will aproximately correspont to the shap of a cell at this radius. To be consistent with axis further, we take points between -pi/7 + pi/2 and pi/7 + pi/2

Here is the plot of the experimental data and the numerical data (for D = 3e7 (introduire la variable)), at the same points (both are averages over time and over the angle $\theta$).

potentiellement en annexe

Concretely, to average the data over the angle $\theta$, we need to modify the interpolation function in CONES. Instead of read coords, the interpolation function read the data in the `coords_seventh.txt` file.

Then, it computes the interpolation and average the data over the angle $\theta$ (between -pi/7 and pi/7), by doing a somme over $\theta$ and dividing by the number of points (150). To go back from 150x5x3x3 (to detail) to 5x3x3 data. All of those computation are mad in a cylindrical referencial. Ref au schéma tikz

A sixth coefficients DP has been add. It's represent the difference of pressure between the inlet of the pipe and the outlet of the diffuser. The numerical coefficient are interpolating, respectively at (0,0,-0.35) and (0,0.390,0). For the high fidelity value, we can find experimental data form the same pump [M. Fan 2022], that's match with M. Fan simulation (eq to approximately $\eta = 0.4$). So can also compute :

U_2 is the impeller tip speed computed from the impeller diameter and the rotation speed of the pump.

$$\frac{\Delta P_P}{\rho U_2^2} = \eta \Leftrightarrow \Delta P_P = \rho U_2^2 \eta \tag{2.19}$$

U2carré = (omega * R2)² = (2 * pi * 1200 / 60 * 0.2566)² = 32,2 m/s

As we are in a steady simulation, the restult of the numerical pressure is divided by the density of the fluid.

$$\Delta P_P = 32.2^2 \times 0.4 = 415 Pa \tag{2.20}$$

So the state vector is now 46. Incertitude on P will be less to take more take it account. Inflation...

At pump flow rate = pump design flow rate So Q?/Q? =1

### 2.4.2 Modifications on OpenFoam and CONES

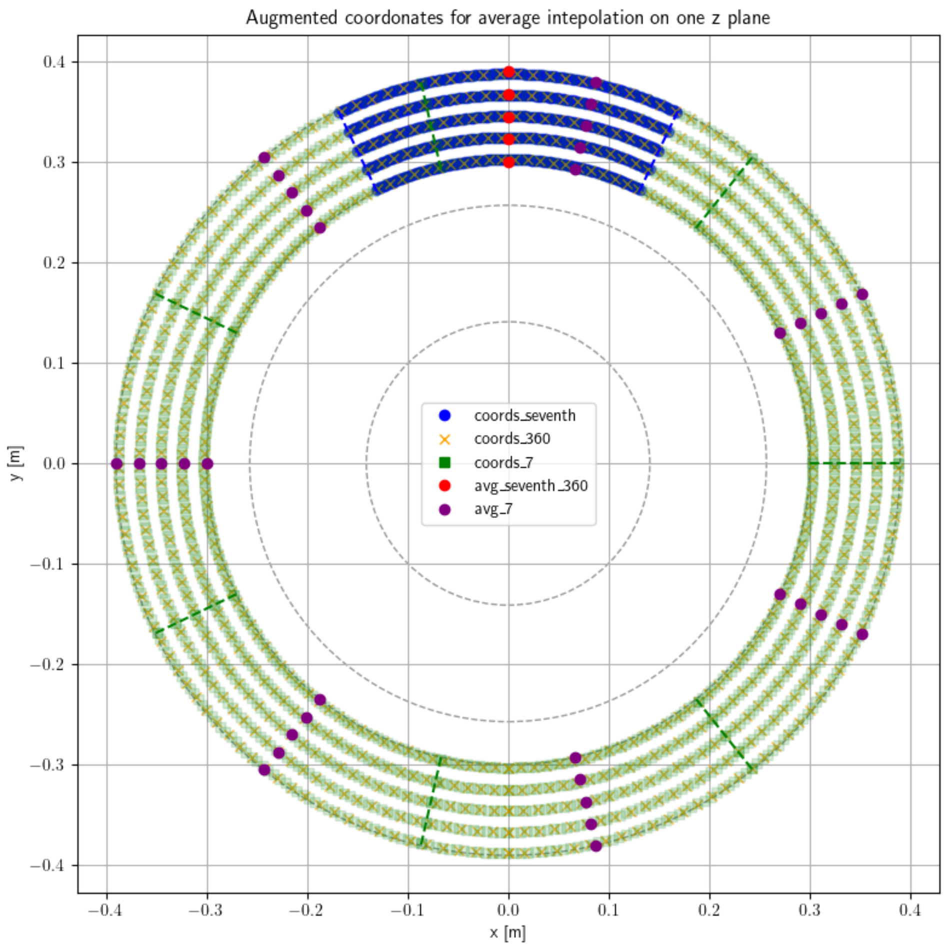



Figure 2.21

Tout ce qu'il y a dans le CONES dict Parler des modifs dans le code ici.

- Reduction of the pipe (with the research of the velocity inlet condition)

Toujours mentionner les paramètres utilisés (dans le controlDict et le CONESDict)

### 2.4.3 Application of the DA

## 2.5 Results

Comparer les 2-3 résultats : (avant DA, déjà fait section précédente), après simple et "complex" DA

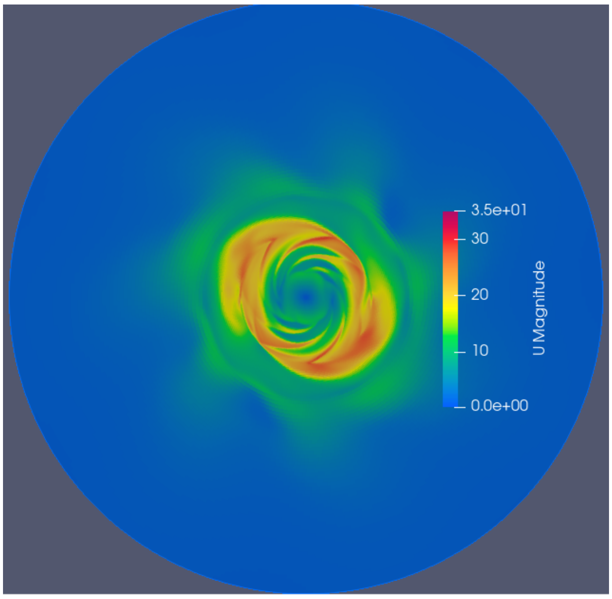



Figure 2.22: Velocity field for Dx = 4e7 and Dy = 2e7

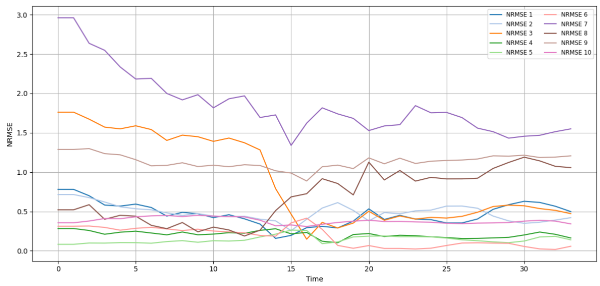



Figure 2.23: Evolution of the NRMSEs over EnKF analysis phases

...

Explication des écarts par différentes choses

Tt ce que j'ai implémenté en plus (déjà détaillé partie précédente normalement)

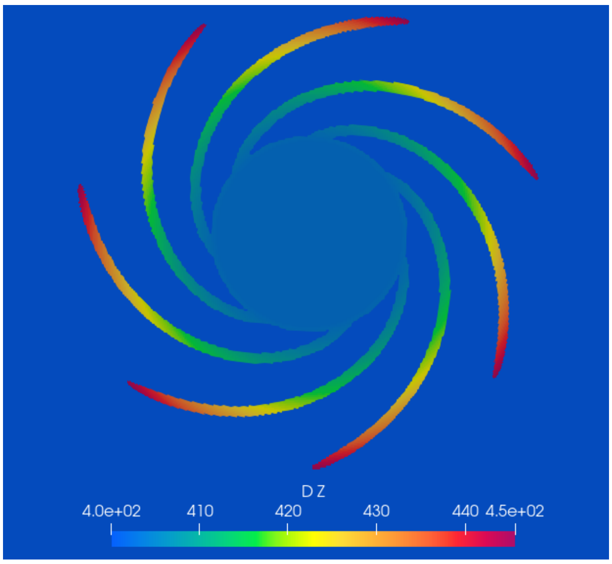



Figure 2.24: Radial expension of Dz over the impeller radius

Essayer d'améliorer le rendu de l'image ci-dessous en mettant la pompe en transparence

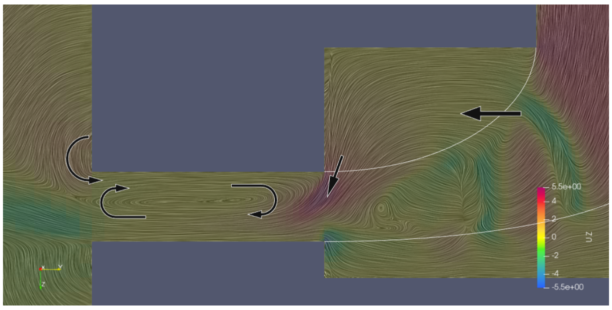


Figure 2.25: Formation of the recirculation bubble in the diffuser

Explication des écarts par la bulle de recirculation.

### 2.5.1 Conclusion ans perspective

Problème de méthode et pas de miracle ave cette méthode IBM. Interpolation, instationnaire (avantage : on ne contraint plus avec MRF et donc plus facile de réconcilier solution numérique physique et mathématique)

Ouverture : méthode IBM interpolation, article non référencé.

[9] -> ML turbulent channel flow Miguel

"Ouverture / idée" : DA temps réel dans avion impossible pour encore longtemps. Solution : Contrôle actif de la puissance en fonction des mesures de pression dans le compresseur : f(obs) -> controle de la puissance Pour ça, se ref au sujet de thèse de Tarane obs pression –> modèle de reconstruction du champ de vitesse –> modèle de reconaissance de patternes de décrochage –> contrôle de la puissance Peut-être faire deux filtres de Kalman imbriqués ? hyperparamètres type fenetre observation, variance initiale, ...

Voir "TD -> Done et "TDSave -> Done" pour voir tt ce que j'ai fais

The random variations are random but with a reproducibility seed (sed) that allows to have the same result each time the simulation is launched.

//The theorical average radial speed at the input of the diffuseur is equal to

$$\frac{Q_{dif_{in}}}{S_{dif_{in}}} = \frac{Q_{pump_{in}}}{2\pi R_3 h_{dif}} = 3.65 m.s^{-1}$$

With $Q_{pump_{in}} = Q_{dif_{in}}$ because the mass flow rate is constant and the flow is incompressible.// -> No need ??

Parler du cas test profil d'aile de Tom

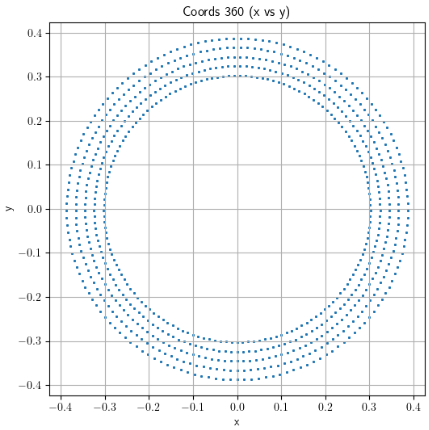



Figure 2.26: Interpolation points over 360 degrees

# Conclusion

### Technical and human review

Véirifier les consignes

### Outlook

### Personal reflection

# Annexes

## Annex 1 – Batch code

```
#!/bin/bash                     # To be run with bash
#MSUB -r LCP35                  # Job name
#MSUB -o LCP35_%I.out           # Output name. %I is the job ID
#MSUB -e LCP35_%I.err           # Error name. %I is the job ID
#MSUB -q rome                   # Partition
#MSUB -A gen1741                # Project ID
#MSUB -n 4446                   # Cores nb (35 members x 127 OF) + 1 python
##MSUB -c 2                     # To allocate 2 cores per task (default is 1)
##MSUB -N 1                     # Nb of calcul node (128 cores per node on rome)
#MSUB -T  43200                 # Elapsed time limit in seconds
##MSUB -Q long                  # (upto 3 days)
##MSUB -Q test                  # (short, high priority jobs)
#MSUB -m scratch                # Excecute in scratch directory
#MSUB -@ name@mail.fr:begin,end # Email address

set -x
cd ${BRIDGE_MSUB_PWD}
module purge
module load gnu/8 mpi/openmpi/4 flavor/python3/cuda python3/3.10.6 openfoam/9
source $FOAM_SOURCE_FILE

./visuNParams.sh                # Script .sh
ccc_mprun -f app_mySF35.conf    # Run the application with the configuration file
```

# Annex 2 – OpenFOAM Directory Tree

```
root
├── 0
│   ├── k
│   ├── omega
│   ├── p
│   └── U
├── 0.orig
│   └── 0
├── system
│   ├── controlDict
│   ├── blockMeshDict
│   ├── decomposeParDict
│   ├── fvSchemes
│   └── fvSolution
├── constant
│   ├── momentumTransport
│   ├── transportProperties
│   └── MRFProperties
├── dummy.foam
├── processor0
├── processor...
└── processorN-1
```

# Bibliography


[1] Alexis Boudin. "Numerical Methods for Large Eddy Simulation in Turbomachinery (in progress)". PhD thesis.

[2] Mark Asch, Marc Bocquet, and Maëlle Nodet. *Data Assimilation: Methods, Algorithms, and Applications.* Fundamentals of Algorithms 11. Philadelphia: Society for Industrial and Applied Mathematics, 2016. 1 p. ISBN: 978-1-61197-454-6.

[3] A. Dazin et al. "High-Speed Stereoscopic PIV Study of Rotating Instabilities in a Radial Vaneless Diffuser". In: *Experiments in Fluids* 51.1 (July 2011), pp. 83–93. ISSN: 0723-4864, 1432-1114. DOI: 10.1007/s00348-010-1030-x. URL: http://link.springer.com/10.1007/s00348-010-1030-x (visited on 04/18/2025).

[4] Meng Fan et al. "Effect of Leakage on the Performance of the Vaneless Diffuser of a Centrifugal Pump Model". In: (2022).

[5] Tom Moussie, Paolo Errante, and Marcello Meldi. "Statistical Inference of Upstream Turbulence Intensity for the Flow Around a Bluff Body with Massive Separation". In: *Flow, Turbulence and Combustion* 113.4 (Nov. 2024), pp. 853–889. ISSN: 1386-6184, 1573-1987. DOI: 10.1007/s10494-024-00573-z. URL: https://link.springer.com/10.1007/s10494-024-00573-z (visited on 05/06/2025).

[6] Charles S Peskin. "Flow Patterns Around Heart Valves: A Numerical Method". In: *Journal of Computational Physics* (1972).

[7] Ladislao Reti and Francesco Di Giorgio Martini. "Francesco Di Giorgio Martini's Treatise on Engineering and Its Plagiarists". In: *Technology and Culture* 4.3 (Sum. 1963), p. 287. ISSN: 0040165X. DOI: 10.2307/3100858. JSTOR: 3100858. URL: https://www.jstor.org/stable/3100858?origin=crossref (visited on 06/26/2025).

[8] M. M. Valero and M. Meldi. "An immersed boundary method using online sequential data assimilation". In: *Journal of Computational Physics* 524 (2025), p. 113697.

[9] Miguel M. Valero and Marcello Meldi. *Enhanced State Estimation for Turbulent Flows Combining Ensemble Data Assimilation and Machine Learning.* Jan. 30, 2025. DOI: 10.48550/arXiv.2501.18262. arXiv: 2501.18262 [physics]. URL: http://arxiv.org/abs/2501.18262 (visited on 07/09/2025). Pre-published.

[10] Lucas Villanueva et al. *Augmented State Estimation of Urban Settings Using Intrusive Sequential Data Assimilation.* Jan. 26, 2023. DOI: 10.48550/arXiv.2301.11195. arXiv: 2301.11195 [physics]. URL: http://arxiv.org/abs/2301.11195 (visited on 07/08/2025). Pre-published.

# List of acronyms

**AI** Artificial Intelligence

**CERFACS** Centre Européen de Recherche et de Formation Avancée en Calcul Scientifique

**CFD** Computation Fluid Dynamics

**CFL** Courant-Friedrichs-Lewy

**CHSCTN** Comité d'Hygiène, de Sécurité et des Conditions de Travail National

**CNRS** Centre National de la Recherche Scientifique

**CONES** Coupling OpenFoam with Numerical EnvironmentS

**CPER** Contrat de Plan État-Région

**CSE** Comité Social et Économique

**CSR** Corporate Social Responsibility

**CTI** Commission des Titres d'Ingénieur

**DA** Data Assimilation

**DNS** Direct Numerical Simulation

**EnKF** Ensemble Kalman Filter

**ENSAM** École Nationale Nupérieure d'Arts et Métiers

**HCERES** Haut Conseil de l'Évaluation de la Recherche et de l'Enseignement Supérieur

**IBM** Immersed Boundary Methods

**IPSA** Institut Polytechnique des Sciences Avancées

**JFM** Journal of Fluid Mechanics

**LES** Large Eddy Simulation

**LMFL** Laboratoire de Mécanique des Fluides de Lille

**MPI** Message Passing Interface

**MRF** Multiple Reference Frame

**ONERA** Office National d'Études et de Recherche Aérospatiales

**OpenFOAM** Open source Field Operation And Manipulation

**PIV** Particle Image Velocimetry

**RANS** Reynolds-Averaged Navier–Stokes equations

**RSE** Responsabilité Sociétale des Entreprises

**RSU** Rapport Social Unique

**SHF** Société Hydrotechnique de France

**SST** Shear Stress Transport

**Résumé** Vérifier les consignes

**Abstract**